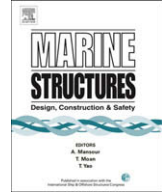




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Derivation of wave spectrum using data driven methods

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ARTICLE INFO

Article history:

Received 30 July 2008

Received in revised form 21 October 2008

Accepted 5 December 2008

Keywords:

Support vector regression

Model trees

Wave spectra

Wave measurements

ABSTRACT

The current techniques of derivation of a wave spectrum from given values of design wave parameters, like significant wave height and average wave period, are fraught with considerable uncertainties. This leaves scope for alternative approaches. The reported work proposes potential applications of two recent data driven methods, namely support vector regression (SVR) and model tree (MT), to obtain the wave spectra. In the present study the above tools were used to estimate wave spectra at two locations: no. 44008 maintained by National Data Buoy Centre (NDBC) in the Gulf of Maine, USA and 'DS5' monitored by National Institute of Ocean Technology (NIOT) in Bay of Bengal, India. The choice of these two locations facilitated the comparison of model performances in different geographical areas. The SVR and MT models were developed in order to estimate the wave surface spectral density over a wide range of wave frequencies out of average wave parameters of significant wave height and average zero-cross wave period. The models were trained and tested using randomly selected sea states. Both MT and SVR were able to derive the spectral shapes satisfactorily as reflected in high values of the correlation coefficients and low values of root mean square error and mean square error.

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1. Introduction

A wave spectrum represents the distribution of wave energy over various wave frequencies. Such a spectrum has number of applications in structural analysis and design, studies on coastal processes,

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wave prediction models, simulation of random waves in a laboratory and so on. Many times the wave spectra are required to be estimated from given values of average wave parameters, namely, the significant wave height, H_s and the average zero-cross wave period, T_z only and in such cases the use of the empirical or theoretical spectral relations is common. The resulting formulations include Pierson Muskowitz (PM), JONSWAP, Scotts and others [1]. Although the use of these spectra is common in design of coastal and offshore structures, studies in the past [2–4] have cautioned against their use without proper verification owing to a large number of uncertainties associated in determining the spectral distribution. Such uncertainties in the estimation of spectral shapes can arise out of two factors: one is the randomness of the wave phenomenon in general and the second is the limitation of the method of analysis employed to analyze observed data, which had been mostly the dimensional analysis and non-linear regression. Recent advances in computer technology have given rise to new tools of data analysis and modeling, e.g., the artificial neural network (ANN), which has been proved to perform better than the statistical regression in many applications such as water resources [5] and ocean engineering in general [6]. The ANN works on the same lines as that of the human brain in following any cognition process. Refs. [7,8] applied the technique of ANN to estimate the wave spectra for specified H_s and T_z values at a few locations of the US and Indian coasts and noticed more accurate results than the theoretical spectra.

Although the above mentioned attempts met with good success the problem of deriving accurate spectral distribution for given values of design significant wave height and average zero-cross wave period still involved considerable uncertainty and hence remained elusive going by the accuracy and generality of the results reported in these publications. An exploration of alternative and more recent data driven models for this purpose should therefore be welcome. Recently the use of two statistical learning methods, namely support vector regression and model tree have been successfully applied by researchers in problems such as lake water level predictions [9] and rainfall – runoff modeling [10], respectively. These techniques can map a given random input vector with corresponding output vector by carrying out a piecewise or local learning and appear to be fit to apply to the problem under consideration, namely, producing the spectral energy distribution over different wave frequencies out of given average wave parameters. The present study addresses this issue.

2. A brief overview of the methods used

2.1. Support vector regression

Support vector regression (SVR) is borne out of support vector machine (SVM), which is a type of machine learning techniques. It derives its name because of a set of training pairs called support vectors which are specifically selected for model building. SVM is essentially a classification tool, but after modifications to its optimization function it can be used as a regression scheme and in that case it is better known as support vector regression. Such modification also involves introduction of an appropriate distance metric. The theoretical details of SVR are explained in [Appendix A](#). In SVR the data belonging to a given input space are transformed into a feature space and within this feature space linear models are built up based on the selection of a part of data as per the distance metric. The transformation is achieved using Kernel functions of say Gaussian or polynomial type. Basically SVR optimally separates two or more classes in data by finding a demarcating hyperplane that maximizes the margin, or the highest distance between two closest points on both sides of the hyperplane (see [Fig. 1](#)) and also minimizes the error of misclassification ' ξ '. Vapnik [11] innovatively combined the existing basic ideas such as hyperplanes, margins, kernels and came up with the SVM formulation having a high capacity to learn on its own [12,13].

SVR is analogous to more traditional method of ANN in functioning but differs in the training scheme. It always converges to an optimal solution while learning and comes up with a fixed outcome each time. It uses the structural risk minimization concept to select training parameters, which involves consideration of an expected risk and placing an upper limit on it. This guarantees a unique global minimum of the error surface through solution of a quadratic optimization function. The SVR, unlike ANN does not involve any trial based approach to select control parameters as the algorithm itself produces their values.

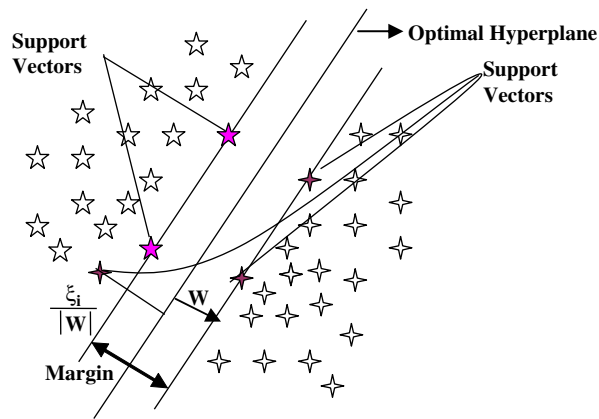


Fig. 1. Support vector regression - optimal hyperplane separating two data classes.

2.1.1. Past works in water related areas

SVM or SVR has been applied by a few investigators to solve problems in water resources [14–21] but not so in ocean engineering. The performance of SVR with other techniques had been assessed by some investigators. Dibike et al. [14] found SVM to perform better than self-organized feature maps and conceptual ANN models and also noticed that the RBF Kernel worked better than polynomial one. Singh et al. [21] also found the RBF Kernel better than the polynomial one. Vojinovic and Kecman [15] cautioned that in SVR also an overfitting may occur like ANN, if support vectors become large in number compared to the training examples. Cannas et al. [16] observed that if data non-stationarity is very high no single model (SVM, multi layer perceptron (MLP), self-organizing maps (SOM), auto-regressive moving average (ARMA)) works satisfactorily and instead the MLP trained by an efficient data pre-processing is sufficient. Sivapragasam and Muttill [17] found SVR to be better than higher order polynomial and logarithmic methods. Khan and Coulabay [9] SVR model outperformed the common ANN and seasonal auto-Regressive (AR) methods. The better generalization by SVM achieved because of the optimization process followed by it (that puts an upper limit on expected risk rather than doing a reduction in training data error) seemed to be the reason for SVM's better performance. An et al. [13] found SVM to work slightly better than the discriminant analysis in their study and also opined that the degree of the polynomial Kernel might not be crucial.

2.2. Model tree

The concept of model tree is based on extending that of the classification tree and the regression tree, in which the computational process is represented by a tree structure (Fig. 2). Such a structure

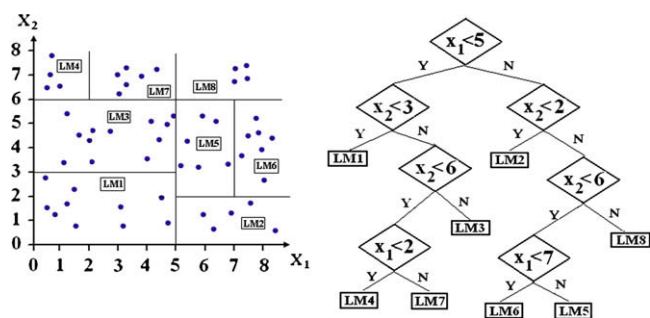


Fig. 2. The model Tree (Y = Yes; N = No).

consists of a root node or a decision box branching out to numerous other nodes and leaves. In a model tree the entire input or parameter domain is divided into sub-domains and a multiple linear regression model is developed for each sub-domain. It therefore formulates a piecewise linear model to approximate the given non-linear relationship between a dependent variable and corresponding independent variables. There are alternative algorithms to build MT and these depend on the domain splitting criterion employed. The domain splitting is done using the concept of maximization of entropy or minimization of uncertainties. There are thus methods based on spline or standard deviation reduction to divide the input domain. A method called the 'M5' algorithm given by Quinlan [22] is popular. In this algorithm the domain splitting is done with the help of the standard deviation of the class value reaching a node. Fig. 2 shows a tree structure representation of the training process and corresponding to a given 2-D input domain of parameters x_1 and x_2 . The difference between the standard deviation of the total training set and that of any i -th sub-domain is taken as the splitting criterion, after weighting the latter. Typically we calculate:

$$\sigma_R = \sigma(N) - \sum_{i=1}^N \left(\frac{N_i}{N} \right) \cdot \sigma(N_i) \quad (1)$$

Where σ_R = standard deviation reduction, $\sigma(N)$ = standard deviation of all training samples having a total number, N ; N_i = i -th sub-domain of N , $\sigma(N_i)$ = standard deviation of the i -th sub-domain. Many possible ways of input divisions are examined and finally the one that ensures the maximum value of standard deviation reduction is selected to build linear models in each sub-domain. This model could be typically:

$$o = a_0 + a_1x_1 + a_2x_2 + \dots \quad (2)$$

Where o = output value; $a_0, a_1, a_2, \dots$ = coefficients derived using say a least squares fit; $x_1, x_2, \dots$ = input values.

The splitting stops when the class values of all collections in a sub-domain do not vary much, say by 5%, or when very few collections are left out. To begin with an attribute, say, $x_1 < 5$ (Fig. 2) is selected as the root node and two branches are drawn (as per 'Yes' or 'No') and thereafter such a process of splitting continues. The selection of the attribute as root node is done as per its ability to maximize the standard deviation reduction.

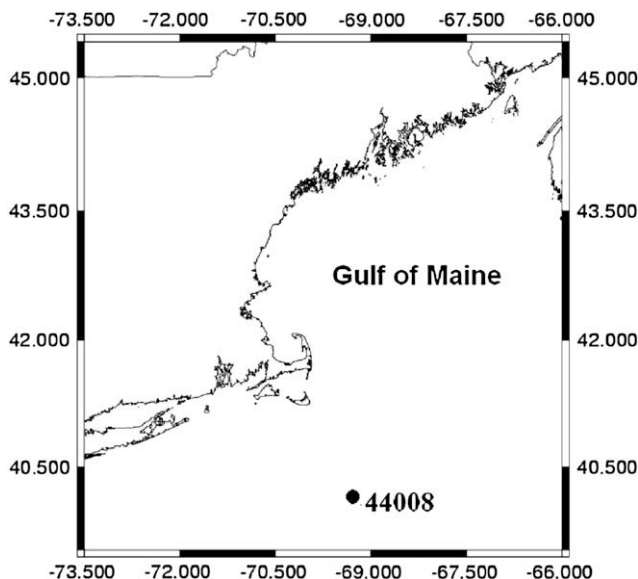


Fig. 3. The US South-East coastline showing station 44008.

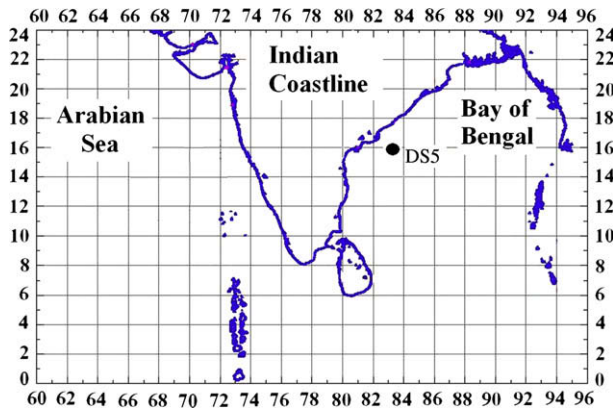


Fig. 4. The Indian coastline showing station DS5 (www.niot.res.in).

The splits are most of the times so large that an overfitting might happen, in which case the generalization does not take place and instead individual training patterns only are learnt. To overcome this, ‘pruning’ of the structure is done by say merging together a few lower sub-domains producing similar models. There is also a possibility that the above pruned structure might have large discontinuities between neighboring models (especially when the training examples are less). Therefore ‘smoothing’ is done by revising nearby equations such that their output comes more close to each other [24].

An advantage of MT over other data mining approaches like ANN or program based genetic programming (GP) is its portability in that its outcome is understandable and can be easily applied by another user. However it is to be noted that being a piecewise linear model it is not purely non-linear like ANN or GP.

2.2.1. Past works on model trees

Applications of MT to solve problems in ocean engineering are yet to be reported in good number. There are however considerable ones in the field of water resources and hydraulics as described in Refs. [10,23,25–30]. Comparisons of MT with artificial neural networks made by these authors showed varying results. While Bhattacharya and Solomatine [26] report noticeably better accuracy of MT, Solomatine and Xue [23] and Bhattacharya et al. [27] found MT only marginally better than or showing similar performance like ANN.

3. The database used

For the present study datasets pertaining to two different locations were used. The first one was station no. 44008 (40.5 N, 69.43 W) maintained by the US National Data Buoy Center (NDBC). This

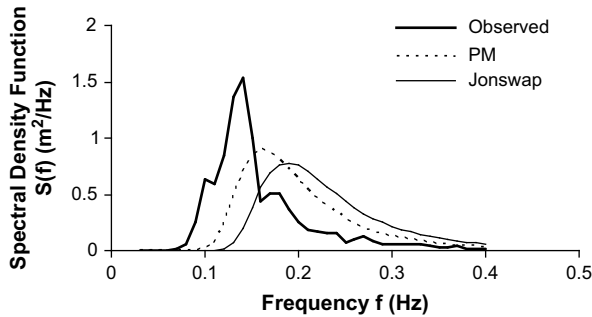


Fig. 5. Comparison of wave spectra at 04 H on 26/07/2003 (NDBC).

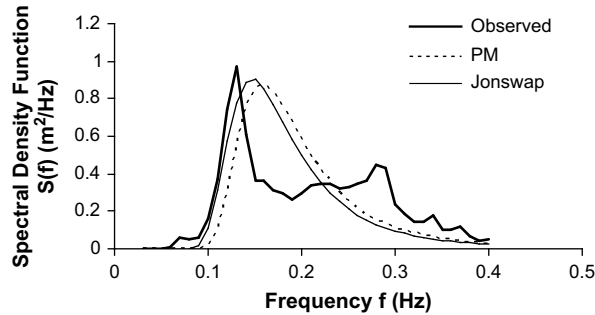


Fig. 6. Comparison of wave spectra at 19 H on 16/11/2003 (NDBC).

location is of South-East US coast and it is located at 54 Nm south of south-east of Nantucket in the Gulf of Maine (see Fig. 3). The wave measurements were made by a 10 m DISCUS buoy in 62.5 m depth of water. The hourly measurement of spectral density, $S(f)$ with varying frequencies, f , significant wave height, H_s and average time period, T_z was downloaded from website: www.ndbc.noaa.gov for a period five years: 1999–2003. The recording frequency varied from 0.03 Hz to 0.40 Hz. The observations were made over an interval of 0.001 Hz. In total 550 time histories were randomly selected for this study during the five year period out of which 480 were used for model building and remaining for testing. During the observation period the significant wave height ranged from 0.18 m to 3.69 m with a mean value of 1.15 m. Similarly the average zero-cross period varied from 2.16 s to 8.94 s having a mean of 5.12 s.

The second site was a deep water station DS5 (14°01'28" N, 83°17'26" W) in the Bay of Bengal along the east coast of India (Fig. 4). The measurements made by a DISCUS type of buoy by National Institute of Ocean Technology (NIOT) were used in this study. The available database in this case was smaller than the earlier site and consisted of measurements of March and April, 2003. For this site 3-hourly measurements of spectral density functions along with corresponding frequencies, significant wave height and average wave period were considered. There were in total 240 time histories out of which 200 time histories were used for training and remaining 40 for testing. During the period of observation considered the significant wave height ranged from 0.47 m to 1.95 m with a mean value of 0.89 m. Similarly the average zero-cross period varied from 2.43 s to 9.63 s having a mean of 6.74 s.

The models of SVR and MT were built on the training set of data and were validated for the test samples. In the case of NDBC measurements it was found necessary to develop two separate models for higher and lower sea states in order to impart proper training. This was due to a large range variation in sea state magnitudes, which may make the learning process more complex for a single model. The dividing sea state was arrived at empirically and this corresponded roughly to the mean value of the significant wave height.

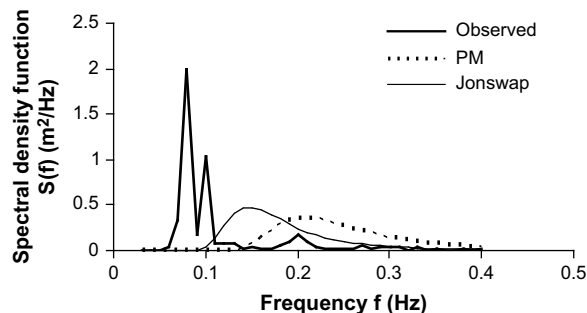


Fig. 7. Comparison of wave spectra at 18 H on 19/04/2003 (NIOT).

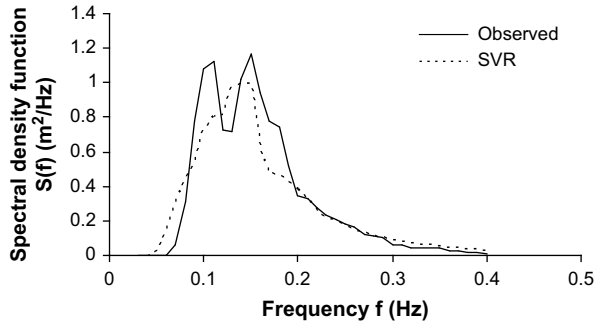


Fig. 8. Comparison of wave spectra at 06 H on 14/04/2003 (NDBC) ($H_s = 1.41$ m; $T_z = 5.97$ s).

To develop the SVR model for given training samples, one has to formulate the objective function: $Q(\cdot)$ using equation [11] of Appendix A and during the application phase obtain the Lagrange multipliers: α_i and α'_i and with their help evaluate output y for given query point $\mathbf{x}_i$. For the MT, the application procedure would involve selecting a root node randomly and carrying out the splits with the help of equation (1) and stopping when certain minimum number of instances is left out in each leaf. Linear multiple regression models are to be developed for each leaf thereafter. After such calibration and for a new query an appropriate model is to be selected as per the input values and the corresponding output is to be obtained accordingly.

The closeness of predictions with actual observations was judged qualitatively by plotting a graph of the spectral density function against various wave frequency values and quantitatively by evaluating three error measures, namely, correlation coefficient, R , root mean square error, RMSE and mean absolute error, MAE.

4. Results and discussion

The objective of this study is to explore usefulness of the data driven methods of SVR and MT in deriving the wave spectrum for specified input wave parameters characterizing the short term sea state in an average manner. The latter values normally available to a designer are in the form of significant wave height, H_s and average zero-cross period T_z defined below [1].

$$H_s = 4\sqrt{m_0} \quad \text{and} \quad T_z = \sqrt{\left(\frac{m_0}{m_2}\right)} \quad (3a)$$

where

$$m_0 = \int S(f)df \quad \text{and} \quad m_2 = \int f^2 S(f)df \quad (3b)$$

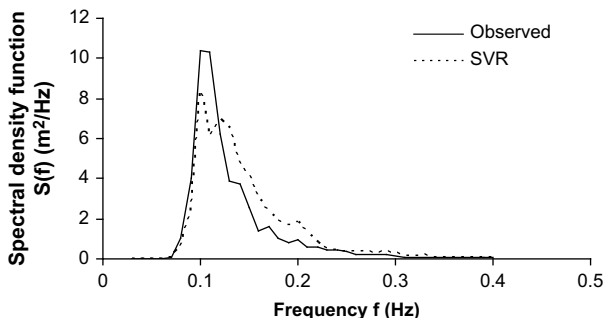


Fig. 9. Comparison of wave spectra at 08 H on 22/10/2003 (NDBC) ($H_s = 2.88$ m; $T_z = 7.14$ s).

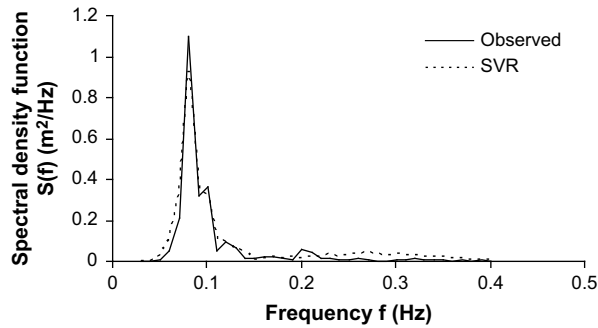


Fig. 10. Comparison of wave spectra at 12 H on 20/04/2005 (NIOT) ($H_s = 1.02$ m, $T_z = 5.86$ s).

Since 1940s it was believed that the high frequency part of a wave spectrum had a certain fixed shape indicating a state of equilibrium with wind input. For such a condition the wave energy density is proportional to f^{-m} , where m might vary from 4 to 6 depending on a variety of conditions such as deep or shallow water, presence or absence of swell, dominance of refraction, sea bed friction or breaking, wind duration and fetch restrictions [31–33]. Details of corresponding physical processes can be read in Kinsman [34], Shore Protection Manual [35] and Young [36]. Taking a cue from such observations theoretical empirical equations relating $S(f)$ to ' f ' were presented by Pierson and Muskowitz, Darbyshire, Sockts and Hasselmann et al. [1].

In the case of present data a look at the true spectra derived for both the locations under consideration showed that most of the spectral energy was concentrated within the frequency range of 0.05–0.40 Hz. However depending on local met-ocean processes the peaks of the spectra occurred at different frequencies. For the NDBC location such peaks were located at around 0.12 Hz while for the NIOT site the same were stationed at around 0.09 Hz. The spectral peak was not found to shift to other frequencies when the sea state intensity or similar conditions changed. Some times presence of a distinct secondary peak was seen for some spectra when the energy levels were falling. Such multi-modal spectra are commonly observed, e.g., see Narasimhan and Deo [37]. When the primary and secondary peaks occur such that they can be interrelated in terms of frequencies it indicates that they belong to the same wave system and arise due to the non-linear mechanism of wind wave generation that excites higher harmonics [31]. Multiple peaks also occur when locally generated waves are superposed on swells arriving from a distant storm or when new wind waves are created before subsidence of a previous storm. They in general represent combined sea and swell conditions. In the present case for the NDBC location secondary peaks were not always seen to occur at a fixed frequency related to that of the peak frequency but at varying frequencies of 0.16–0.28 Hz (referred to Figs. 5, 6, 8,

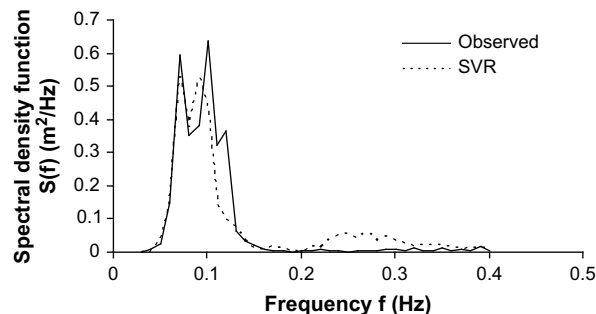


Fig. 11. Comparison of wave spectra at 06 H on 25/04/2005 (NIOT) ($H_s = 0.94$ m, $T_z = 7.27$ s).

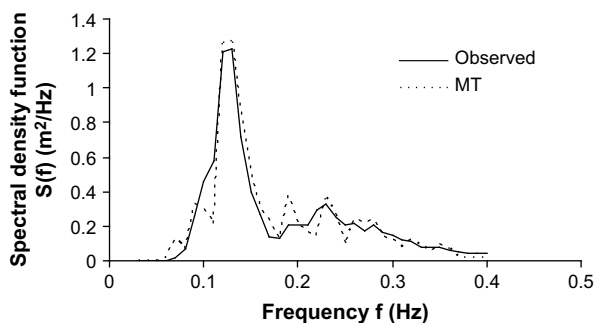


Fig. 12. Comparison of wave spectra at 9 H on 29/4/2003 (NDBC) ($H_s = 1.19$ m, $T_z = 5.19$ s).

12, 13 and 15 quoted later). In the case of the NIOT site occurrence of secondary peaks was not so distinct and common, and unlike the NDBC location they occurred at a very close interval to that of the peak frequency (see Figs. 7, 11, and 17 referred to later).

Before considering application of the data driven tools it was thought worthy to examine how far the commonly used existing empirical spectra represent the observed spectral shapes. The theoretical spectra considered were PM and JONSWAP evaluated as per equations given in Appendix B.

Figs. 5–7 show typical comparisons of observed spectra with the PM and JONSWAP formulations for the two sites under consideration. It may be seen that there is a considerable difference between the PM, JONSWAP and the observed spectral shapes. As can be seen in these examples there was a shift in the occurrence of peak frequency, the peaks were not predicted properly and the shape of spectral curve was also incorrectly estimated. The theoretical spectra were based on a traditional statistical analysis involving curve fitting based on the assumption of a fixed form of input–output relationship a priori and thus had less flexibility in model building compared with the current techniques that do not make such assumptions. Further the theoretical spectra were validated with respect to the North Sea observations in which the met-ocean conditions could be different than those prevailing elsewhere.

The above graphs are indicative of inadequacy of the theoretical spectra in approximating the true spectral shapes. In the subsequent stage therefore the same task was targeted through the use of selected data driven tools.

While using SVR and MT a single model for all wave frequencies was found to be difficult to obtain owing to the complexity of algorithms that made only a single output node as mandatory. The method therefore involved the use of two inputs namely H_s and T_z and only one output of $S(f)$ at a given frequency f . The magnitudes of spectral density functions beyond the range of 0.0–0.4 Hz were found to

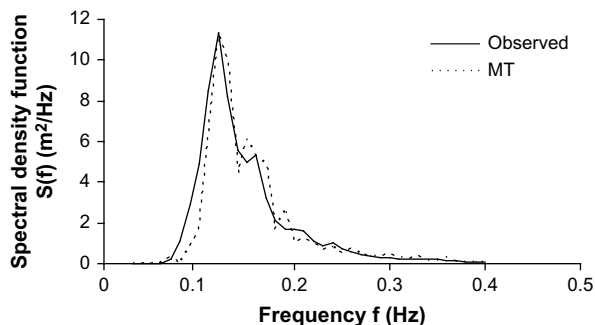


Fig. 13. Comparison of wave spectra at 15 H on 16/12/2003 (NDBC) ($H_s = 3.36$ m, $T_z = 6.35$ s).

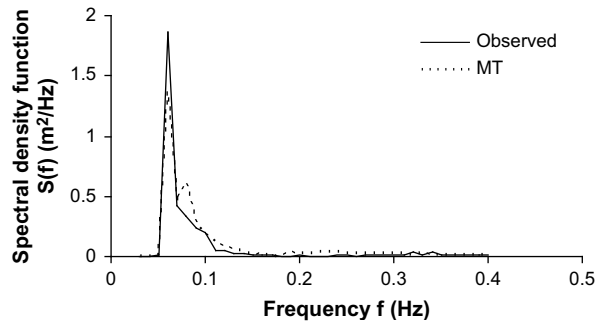


Fig. 14. Comparison of wave spectra at 18 H on 22/04/2005 (NIOT) ($H_s = 0.78$ m, $T_z = 5.51$ s).

be negligible. Hence this range of frequencies only was considered and divided into 38 values equally and for each one of them the spectral densities were estimated. The figure of 38 was arrived at by trial and as a trade off between the conflicting requirements of preserving the details of a given sea state and obtaining a smooth spectral curve. The various $S(f)$ values so derived were later on combined to obtain the full wave spectrum.

The application of SVR made under the Matlab environment involved selection of the appropriate kernel function and associated parameters. The Kernel function selected was radial basis function (RBF) since the alternatives of polynomial and linear did not work well in this application. This is in consonance with earlier observations of Refs. [13,14]. The values of the control parameter 'C' (Eq. (19)), which is a regularization parameter that determines the trade off between the fitting error minimization and smoothness of the estimated function was chosen as 10 and that of the Kernel function parameter ' γ ' of the RBF Kernel (Eq. (8)) was 0.3. These values were arrived at by trials performed over a wide range.

Figs. 8–11 show examples of how the SVR yielded spectra compare with the actual observed ones for both stations. It is clear from the results that the SVR model was able to predict the shape of wave spectrum from the short term average parameters of H_s and T_z fairly satisfactorily. However the exact peak density was found to be generally underestimated or overestimated, like most theoretical spectra, but to a smaller extent.

Figs. 12–14 show some of the results obtained by using MT. Similar to the SVR, the MT was also able to yield satisfactory results and in most of the spectra the exact peak density was recognized. When the sea state was high, the data divisions or splits in MT were much larger (typically 20) than the same corresponding to lower sea states. This shows that as expected the number of classifications in MT becomes high with the complexity of data.

For a given short term sample all computed and actual spectral densities were viewed together and one value each of R , RMSE and MAE was derived. An average of these statistics for both the sites is shown in Table 1. Examples of how both SVR and MT models performed for the same observed spectrum are given in Figs. 15–17.

Figs. 15–17 and Table 1 indicate that when all the error statistics were viewed together the estimation of spectral shapes was better at the NIOT site than at the NDBC location and this could be due to

Table 1
Average error statistics.

Site	SVR			MT		
	R	MAE	RMSE	R	MAE	RMSE
NDBC	0.78	0.40	0.78	0.77	0.35	0.91
NIOT	0.78	0.08	0.03	0.74	0.09	0.03

Note: MAE and RMSE are in m^2/Hz .

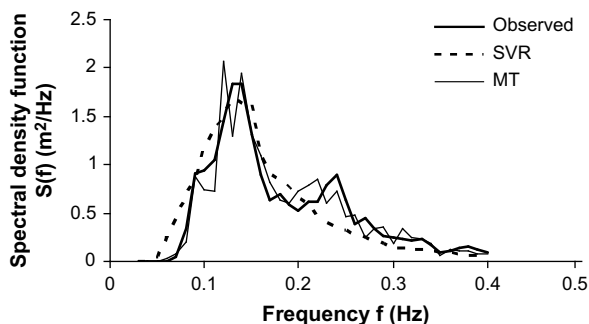


Fig. 15. Comparison of wave spectra at 19 H on 26/05/2003 (NDBC) ($H_s = 1.19$ m, $T_z = 5.19$ s).

the moderate environment prevailing at the former. Also it was noticed that both the methods produced more or less similar results in terms of the prediction accuracy. The treatment to data non-linearity provided by both SVR and MT thus was equally satisfactory.

It has been reported that if the support vectors turn out to be small in number [14] or if the training sample size happens to be small in their comparison [15] then high errors in training can result. Also for low sample sizes a single misclassification can affect accuracy [13]. Cannas et al. [16] pointed out that high non-stationarity of data can make SVR less workable. SVR may also be sensitive to the user defined choice of control parameters. However all these observations were not found to be true here in the light of equally good performance of SVR and MT.

The above results when compared with those based on ANN and on observations at other locations by Naithani and Deo [7] and Namekar and Deo [8] showed lower values of the correlation coefficient in general. However when the other error criteria, namely, mean square error MSE and MAE were taken into account, the present study showed a better model performance. For example, Naithani and Deo [7] achieved MAE of 20–25 m²/Hz and MSE of 4.5–13 m²/Hz and Namekar and Deo [8] had MAE of around 12 m²/Hz and MSE of around 14 m²/Hz. As against this the present study involved MAE and MSE only 0.08–0.40 m²/Hz and 0.009–0.81 m²/Hz for both the sites.

It may be noted that the data driven methods considered in this study have large degrees of freedom associated with them due to which the adequately trained models can estimate different spectral shapes even if the input values that are close to each other are given. Designers and researchers currently use empirical equations to derive shapes of wave spectra from given values of H_s and T_z . Instead, they may consider using new approaches such as those discussed in this paper for more realistically representing observed spectral shapes.

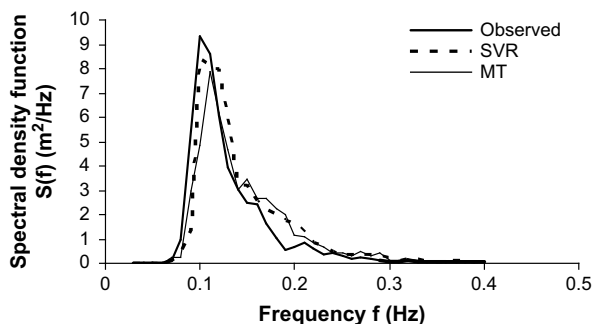


Fig. 16. Comparison of wave spectra at 09 H on 22/10/2003 (NDBC) ($H_s = 3.36$ m, $T_z = 6.35$ s).

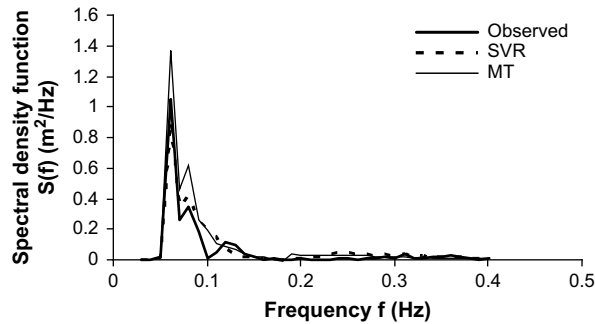


Fig. 17. Comparison of wave spectra at 12 H on 22/04/2005 (NIOT) ($H_s = 0.78$ m, $T_z = 5.51$ s).

5. Conclusions

The current techniques of derivation of wave spectrum from the given values of design wave parameters of significant wave height and average wave period, are fraught with considerable uncertainties. This leaves scope for alternative approaches. The reported work proposes potential applications of two recent data driven methods, namely support vector regression and model tree, to obtain the wave spectra. In the present study the above tools were used to estimate wave spectra at two locations: station no. 44008 maintained by NDBC in the Gulf of Maine, USA and location: DS5 monitored by NIOT in Bay of Bengal. The choice of these two locations facilitated the comparison of models in different geographical areas. The SVR and MT models were developed in order to estimate the wave surface spectral density over a wide range of wave frequencies out of average wave parameters of significant wave height and average zero-cross wave period. The models were trained and tested using randomly selected sea states.

The study once again highlighted importance of developing site specific models to derive wave spectra and also inadequacy of traditionally used empirical theoretical formulations of PM and JONSWAP.

In general both SVR and MT were able to derive the spectral densities satisfactorily for a variety of spectral shapes: single and multi-peaks, narrow and broad banded distribution, high and low energy levels.

The modeling by SVR and MT however was found to be somewhat more satisfactory at the NIOT site where variations in sea states were moderate compared with the NDBC location.

When the sea state was high, the numbers of MT rules were much larger than the same corresponding to lower sea states. This showed that as expected the number of classification in MT became high with the complexity of data.

In general it is felt from this study that the designer can use either SVR or MT for the application; however for getting transparent models the MT can be recommended.

Acknowledgement

The basic MT code used was downloaded from <http://www.cs.waikato.ac.nz/wl/weka>. The basic SVR code belonged to a licensed version of the corresponding Matlab toolbox. Authors thank Dr V. Rajendran and Dr. G. Latha, NIOT, India for sparing data and for helpful suggestions.

Appendix A. Theoretical details of SVM

Vapnik [12] is credited with the development of SVM theory. In SVM modeling, the goal is to classify given variables in two (or more) classes by finding an optimum hyperplane (Fig. 1) that keeps the two types of target variables on two sides of the plane, with a maximum possible distance between them. Such an action is possible by transforming the input space onto a feature space (where the feature is a transformed attribute and attribute refers to a predictor variable). The support vector represents

vectors (or a set of features) nearest to the hyperplane. The mathematical formulation involved in the above process is as below.

An unknown scalar 'd' can be related to a given vector $\mathbf{x}$ through a non-linear function $f(\cdot)$ as follows:

$$d = f(\mathbf{x}) + v \quad (4)$$

where v = independent noise.

If 'y' is the estimate of the true output 'd' then its value for given $\mathbf{x}$ (true input $x_1, x_2, \dots, x_{m_0}$) can be obtained by transferring the input into the feature space (by using kernel functions $K(\cdot)$ indicated later) and linearly regressing such outcomes in the feature space, i.e.:

$$y = \sum_{j=1}^{m_1} w_j \phi_j(\mathbf{x}) + b \quad (5)$$

Where w_j = linear weights connecting feature space to output; b = bias; $\phi_j(\mathbf{x})$ = non-linear transformation function from input space to feature space for true input vector $\mathbf{x}$.

For given sample values $\mathbf{x}_i$ ($i = 1, 2, \dots, N$) of $\mathbf{x}$ the transfer function is related to Kernel functions $K(\cdot)$ as below:

$$K(\mathbf{x}, \mathbf{x}_i) = \Phi^T(\mathbf{x}) \cdot \Phi(\mathbf{x}_i) \quad (6)$$

The Kernel $K(\mathbf{x}, \mathbf{x}_j)$ can be of different types, e.g.,

$$\text{Polynomial : } K(\mathbf{x}, \mathbf{x}_j) = [(\mathbf{x} \cdot \mathbf{x}_j) + 1]^n, \quad (7)$$

$$\text{Radial Basis function : } K(\mathbf{x}, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x} - \mathbf{x}_j\|^2) \quad (8)$$

$$\text{Neural network : } K(\mathbf{x}, \mathbf{x}_j) = \tan h[b(\mathbf{x}, \mathbf{x}_j) - C] \quad (9)$$

where, $n, \gamma, b(\mathbf{x}, \mathbf{x}_j), C$ are user defined parameters.

With the help of the Kernel function the desired estimate 'y' is reformulated as

$$y = \sum_{i=1}^N (\alpha_i - \alpha'_i) K(\mathbf{x}, \mathbf{x}_i) \quad (10)$$

where N = max number of training samples; α_i, α'_i = Lagrange multipliers that maximize the objective function $Q(\cdot)$ given below and thereby achieve generalization.

$$Q(\alpha_i, \alpha'_i) = \sum_{i=1}^N d_i (\alpha_i - \alpha'_i) - \varepsilon \sum_{i=1}^N (\alpha_i + \alpha'_i) - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (\alpha_i - \alpha'_i) (\alpha_j - \alpha'_j) K(\mathbf{x}_i, \mathbf{x}_j) \quad (11)$$

where ε = a user defined parameter.

This optimization is subject to constraint:

$$\sum_{i=1}^N (\alpha_i - \alpha'_i) = 0 \quad (12)$$

with both α_i and α'_i are positive with an upper bound 'C' defined by the user.

The optimization of $Q(\cdot)$ leads to values of α_i and α'_i . When $\alpha_i \neq \alpha'_i$ the corresponding input values are called support vectors and as can be seen from equation (10) these are used to obtain output 'y'.

The optimization function $Q(\cdot)$ given in Eq. (11) was derived as described below and based on minimizing the risk:

$$R = \frac{1}{N} \sum_{i=1}^N L_{\varepsilon}(d_i, y_i) \quad (13)$$

$$\text{subject to } \|w\|^2 \leq C_0, \quad \text{where } C_0 = \text{constant} \quad (14)$$

In Eq. (13) $L_{\varepsilon}(d_i, y_i) = \varepsilon$ - insensitive loss function, so called because if the deviation between the true value and the predicted value, $|d_i - y_i|$ is more than a user prescribed value, ε , the loss function equals such a difference, otherwise it does not exist, i.e.

$$L_{\varepsilon}(d_i, y_i) = |d_i - y_i| - \varepsilon \quad \text{if } |d_i - y_i| \geq \varepsilon \quad (15)$$

$$= 0 \quad \text{otherwise} \quad (16)$$

For solvability (in non-separable cases) the system output is constrained within upper and lower limits by introducing two slack variables. ξ_i and ξ'_i (for $i = 1, 2, \dots, N$) such that

$$d_i - w^T \phi(x_i) \leq \varepsilon + \xi_i \quad (17)$$

$$w^T \cdot \phi(x_i) - d_i \leq \varepsilon + \xi'_i \quad \text{with } \xi_i \text{ and } \xi'_i \geq 0 (i = 1, 2, \dots) \quad (18)$$

In terms of the slack variables the constrained optimization problem reduces to minimizing the cost functional:

$$\phi(w, \xi, \xi') = C \left[\sum_{i=1}^N (\xi_i + \xi'_i) \right] + \frac{1}{2} w^T w \quad (19)$$

The second term arises due to the constraint indicated in Eq. (14). With the help of constant 'C', a regularization parameter supplied by the user, the objective function and the constraints can be combined into the Lagrangian function, J :

$$J(w, \xi, \xi', \alpha, \alpha', \gamma, \gamma') = C \left[\sum_{i=1}^N (\xi_i + \xi'_i) \right] + \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i [w^T \phi(x_i) - d_i + \varepsilon + \xi_i] \\ - \sum_{i=1}^N \alpha'_i [d_i - w^T \phi(x_i) + \varepsilon + \xi'_i] - \sum_{i=1}^N (\gamma_i \xi_i + \gamma'_i \xi'_i) \quad (20)$$

Where, α_i and α'_i = Lagrange multipliers.

The use of γ_i and γ'_i ensure that α_i and α'_i remain as variables. $J(\cdot)$ has to be minimized with respect to w , ξ_i and ξ'_i and maximized with respect to α_i and α'_i , and γ_i and γ'_i . Equating corresponding partial derivatives of the Lagrange function to zero for optimal solutions yields:

$$w = \sum_{i=1}^N (\alpha_i - \alpha'_i) \phi(x_i) \quad (21)$$

$$\gamma = C - \alpha_i \quad (22)$$

$$\gamma'_i = C - \alpha'_i \quad (23)$$

Substituting Eqs. (21–23) into Eq. (20) leads to Eq. (11).

The SVR is an extension of SVM in which the dual problem for regression mentioned below is solved.

For given training patterns: $\{x, d_i\}$, $i = 1, 2, \dots, N$, obtain Lagrange multipliers α_i and α'_i that maximize $Q(\alpha_i \text{ and } \alpha'_i)$ (Eq. (11)) subject to constraints:

$$\sum_{i=1}^N (\alpha_i - \alpha'_i) = 0 \quad (24)$$

and

$$\alpha_i \text{ and } \alpha'_i = (0, C) \quad (25)$$

With these values of α_i and α'_i use equation (21) to determine weight vector $\mathbf{w}$. Those training pairs for which $\alpha_i \neq \alpha'_i$ are called support vectors and these pairs yield the regression 'y' as in equation (5).

Appendix B. The PM and JONSWAP equations

The PM spectrum model is written as [38]

$$S(\omega) = \alpha g^2 \omega^{-5} \exp \left[-0.74 \left(\frac{\omega U_w}{g} \right)^{-4} \right] \quad (26)$$

Where $\alpha = 0.0081$; $g = 9.81 \text{ m/s}^2$; U_w = Characteristic wind speed, which can be related to the significant wave height as follows:

$$U_w = \sqrt{\frac{H_s g \sqrt{\frac{\beta}{\alpha}}}{2}} \quad (27)$$

Where $\alpha = 0.0081$ and $\beta = 0.74$.

The JONSWAP spectrum was developed by investigators led by Hasselman during a Joint North Sea Wave Project. The formula for the JONSWAP spectrum can be written by modifying the PM formulations as follows [38]

$$S(\omega) = \alpha g^2 \omega^{-5} \exp \left[-1.25 \left(\frac{\omega}{\omega_0} \right)^{-4} \right] \gamma^{\exp \left[-\frac{(\omega - \omega_0)^2}{2\tau^2 \omega_0^2} \right]} \quad (28)$$

In which γ = peakedness parameter, and τ = shape parameter. The JONSWAP spectrum, as given in modified form by Goda in 1979 and as reported in Ref. [38] is as below:

$$S(\omega) = \alpha^* H_s^2 \frac{\omega^{-5}}{\omega_0^2} \exp \left[-1.25 \left(\frac{\omega}{\omega_0} \right)^{-4} \right] \gamma \exp \left[-(\omega - \omega_0)^2 (2\tau^2 \omega_0^2) \right] \quad (29)$$

where ω_0 = peak frequency, γ = peak enhancement factor calculated from given H_s and T_z as follows,

$$H_s = (0.11661 + 0.01581\gamma - 0.00065\gamma^2) T_0^2 \quad (30)$$

$$T_0 = (1.49 - 0.102\gamma - 0.00079\gamma^2 - 0.00079\gamma^3) T_z \quad (31)$$

and

$$\tau = \text{shape factor} = \begin{cases} 0.07 & \text{for } \omega \leq \omega_0 \\ 0.09 & \text{for } \omega > \omega_0 \end{cases} \quad (32)$$

$$\alpha^* = \frac{0.0624}{0.230 + 0.0336\gamma - 0.185(1.9 + \gamma)^{-1}} \quad (33)$$

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